

SPECIAL CHARACTERISTICS GUIDE

Identification, Classification, Cascading across APQP Documents | AIAG APQP | Quality Head Agents

WHAT ARE SPECIAL CHARACTERISTICS?

Special characteristics are product or process characteristics for which anticipated variation could significantly affect product safety, regulatory compliance, fit, function, performance, or subsequent processing. They require special controls to prevent or detect failure.

CLASSIFICATION OF SPECIAL CHARACTERISTICS

SYMBOL	NAME	CLASSIFICATION	DEFINITION	CONTROL REQUIREMENT
CC or ♦	Critical Characteristic (CC)	SAFETY / REGULATORY	Could affect safety or non-compliance with 100% inspection or OTRC. With Cpk ≥ 1.67, no inspection plan	
SC or ★	Significant Characteristic (SC)	IMPACT	Could significantly affect fit, function, durability, or appearance. If not satisfied, units must be scrapped. Cpk ≥ 1.33	
■ (varies)	Other Special Char.	CUSTOMER SPECIFIED	Defined differently by each customer. GM requires specific requirements. Toyota may use process chart	

HOW TO IDENTIFY SPECIAL CHARACTERISTICS

- **Customer Drawings and Specifications:** Customer drawings call out CC and SC with specific symbols. Review all callouts on the drawing. Customer may also provide a Special Characteristics List separately.
- **DFMEA Analysis:** During DFMEA, high-severity failure modes (Severity ≥8) linked to safety or regulatory compliance automatically become Special Characteristics candidates.
- **Regulatory Requirements:** Any dimension, parameter, or material property linked to safety regulations (ISO, FMVSS, ECE, IS standards) should be treated as a Critical Characteristic.
- **Customer Feedback / Warranty Data:** Dimensions linked to past warranty failures or customer complaints from similar products should be elevated to Special Characteristic status.
- **Engineering Judgment:** The cross-functional team can designate any characteristic as special based on engineering judgment if failure could have significant consequences.

CASCADING SPECIAL CHARACTERISTICS — APQP DOCUMENT TRAIL

DOCUMENT	HOW SC APPEARS	REQUIRED ACTION
Customer Drawing	CC ♦ or SC ★ symbol on dimension	Identify and list in SC Register — Phase 1
Special Characteristics Register	Full list of all SC/CC characteristics	Create SC Register in Phase 1, update through Phase 4
DFMEA	Severity ≥8 + SC symbol on failure mode	Each CC/SC must have corresponding failure mode in DFMEA
Process Flow Diagram (PFD)	SC symbol on process step where SC is created or controlled	PFD must show which step creates the SC
PFMEA	SC symbol on cause/failure mode, enhanced control plan	PFMEA must have separate row for SC failure modes, SC symbol in control plan
Control Plan (Pre-Launch & Production)	SC symbol in Char. No. column, enhanced frequency	All SC must be in Control Plan with higher inspection frequency, rework instructions
Work Instructions	Highlighted with SC symbol, special attention callouts	Operators must be aware which dimensions are SC
Gauge Plan / MSA	SC gauges required to have GRR <10%	All SC measurement systems must have completed GRR study
Capability Study	SC requires Cpk ≥1.67 (preliminary), ongoing Cpk ≥1.33	SC must meet higher threshold
PPAP Package	SC evidence required in PPAP elements 7, 8, 15, 16	All SC evidence must be included in PPAP submission

COMMON MISTAKES WITH SPECIAL CHARACTERISTICS

- SC identified on drawing but NOT cascaded to PFMEA — most common audit finding

- SC in PFMEA but NOT in Control Plan — fails IATF 8.5.1.1 audit
- SC in Control Plan but GRR not completed for measurement system
- Cpk <1.67 for SC at PPAP — should have been caught in Phase 4
- Operators not trained on which dimensions are SC — reaction plan not followed
- SC list not updated when engineering changes modify the drawing